

# A Nonlinear High-Gain Observer for Systems With Measurement Noise in a Feedback Control Framework

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**Abstract**—We address the problem of state estimation for a class of nonlinear systems with measurement noise in the context of feedback control. It is well-known that high-gain observers are robust against model uncertainty and disturbances, but sensitive to measurement noise when implemented in a feedback loop. This work presents the benefits of a nonlinear-gain structure in the innovation process of the high-gain observer, in order to overcome the tradeoff between fast state reconstruction and measurement noise attenuation. The goal is to generate a larger observer gain during the transient response than in the steady-state response. Thus, by reducing the observer gain after achieving satisfactory state estimates, the effect of noise on the steady-state performance is reduced. Moreover, the nonlinear-gain observer presented in this paper is shown to surpass the system performance achieved when using comparable linear-gain observers. The proof argues boundedness and ultimate boundedness of the closed-loop system under the proposed output feedback.

**Index Terms**—Noise, nonlinear control systems, observers.

## I. INTRODUCTION

HIGH-GAIN observers have developed into an important topic in state estimation and output feedback control of nonlinear systems, beginning with papers such as [2] and [3]. In the absence of measurement noise, this technique robustly estimates the derivatives of the output while achieving fast convergence [2]. Moreover, for a sufficiently high observer gain and a globally bounded controller, the high-gain observer is able to recover the system performance achieved with the state feedback control. Refer to [4] for a survey on high-gain observers.

However, observer theory reveals that a tradeoff exists between the measurement noise sensitivity and the speed of state reconstruction [5]. As the observer gain is increased, the bandwidth of the observer is extended. As the bandwidth increases the high-gain observer asymptotically approaches the behavior of a differentiator, exacerbating the presence of measurement noise. The authors of [6], in the context of discrete-time models,

exploited this knowledge by designing a switched filter composed of two linear filters (one for the transient response and the other for the steady-state response); the value of the estimation error determines which filter is active. The idea is to use a large filter gain (increasing the filter bandwidth) during the transient behavior to elicit a fast recovery of the state estimates. The filter with the smaller gain is active once the estimation error has reached a steady-state threshold, reducing the filter bandwidth and preventing a large magnification of the measurement noise. In [7] the authors seek to minimize the effect quantization error has on shaft encoder measurements by introducing a dead-zone nonlinearity into the state estimation scheme. The dead-zone nonlinearity is used to alternate “smoothly” between varying filter bandwidths to initially achieve fast state estimation and, ultimately, minimize the quantization error.

Recently, others have addressed the issue of measurement noise and observers in [8]–[10]. The work in [8] investigates a high-gain observer with a sign-indefinite gain adaptation for systems with potentially nonlocal Lipschitz functions and noisy output. In [9] the authors propose an extended Kalman filter that utilizes an adaptive high-gain parameter to achieve noise rejection and global convergence of the estimated state when careful tuning is exercised. Naturally, this algorithm requires solving the Riccati equation online. The work in [10] analyzes observers with improved transient performance for both linear and nonlinear systems; the effects of measurement noise are briefly considered for one high-gain observer design. All of these observers incorporate additional features to address complications that arise in the system structure, which increases the complexity during the implementation phase.

The effect of measurement noise in high-gain observers has been studied in [11] and [12]. It is shown in [12] that the steady-state estimation error has a component due to modeling uncertainty, which can be attenuated by increasing the gain. Furthermore, the error has a component due to measurement noise that is amplified by increasing the gain. This tradeoff constrains the observer gain, which reduces the observer’s ability to quickly reconstruct the states. In [12] the authors construct an observer to diminish the manifestations of the tradeoff in the system states. A switched-gain observer is proposed in [12] to force a large gain during the transient period for fast state reconstruction, and allow for a smaller gain once the states are satisfactorily estimated to reduce the effect of noise on the steady-state performance. However, a number of complications are associated with this particular design. The time in which

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the gains are switched, trigger threshold, and system peaking are all design issues that must be addressed.

The purpose of this work is to construct a high-gain observer containing a nonlinear-gain that takes the form of a piecewise linear function with two distinct linear regions. The sections are chosen to correspond to the desired transient and steady-state responses, respectively. By constructing the observer gain in this manner, we can achieve fast state estimation and reduced steady-state error. Furthermore, the observer is devised such that the behavior of the innovation process can be controlled separately from the other system states. This is accomplished by assigning one fast eigenvalue, with the remaining eigenvalues chosen relatively slow. Without this key step, the stability analysis for the proposed observer is unattainable. The analysis provided in this paper focuses on the proposed high-gain observer and the closed-loop system dynamics. The discussion concludes with two simulations. The effectiveness of the proposed nonlinear-gain high-gain observer is evaluated against the complex gain structure in [8]. The final example looks at the system performance when using the linear, nonlinear, and switched high-gain observer designs. In particular, we will demonstrate that the nonlinear-gain observer is sufficient in obtaining the desired estimation error dynamics, while reducing the implementation complexity necessary with the switched-gain observer and the adaptive observer in [8].

## II. PROBLEM FORMULATION AND SYSTEM DESCRIPTION

Consider the nonlinear system

$$\dot{z} = \psi(x, z, \varsigma, u) \quad (1)$$

$$\dot{x} = Ax + B\phi(x, z, \varsigma, u) \quad (2)$$

$$y = Cx + v \quad (3)$$

$$w = \Theta(x, z, \varsigma), \quad (4)$$

where  $z \in \mathbb{R}^l$  and  $x \in \mathbb{R}^n$  are the system states,  $y \in \mathbb{R}$  and  $w \in \mathbb{R}^s$  are the measured outputs,  $u \in \mathbb{R}$  is the control input,  $\varsigma(t) \in \mathbb{R}^p$  represents the exogenous signals and  $v(t) \in \mathbb{R}$  is the measurement noise. The function  $\phi(x, z, \varsigma, u)$  may not be known. The triple  $(A, B, C)$  represents a chain of  $n$  integrators, where it is assumed that  $n \geq 2$ .

The structure for the model (1)–(4) includes mechanical systems, electromechanical systems and systems that can be placed in the normal form satisfying the conditions of input-output linearization [13]. Some examples of mechanical and electromechanical systems where the displacements are measured, but not their derivatives can be found in [14]–[16] for an induction motor, a rotational/translational actuator and a smart material application, respectively. Furthermore, the additional measurement  $w$  may not be needed in every model. In this case, (4) can be removed from the system representation. Yet, many models utilize the extra output. For instance, consider a system in which the dynamics are extended by adding integrators; see [13]. Furthermore, the classic example of the magnetically suspended ball is modeled such that the ball position and current are measurement outputs. The position fits the chain of integrators form (in  $x$ ), whereas the current becomes the state variable  $w$ . Hence, many relevant and interesting systems are encapsulated in the types of models of interest in this work.

*Assumption 1:*

- $\varsigma(t)$  is continuously differential and bounded;
- $\varsigma(t) \in \mathcal{D} \subset \mathbb{R}^p$ , where  $\mathcal{D}$  is compact;
- $v(t)$  is a measurable function of  $t$  and bounded, where the bound is defined as  $|v(t)| \leq \mu$ ;
- $\phi$ ,  $\psi$ , and  $\Theta$  are locally Lipschitz in their arguments, uniformly in  $\varsigma$ , over the domain of interest; that is, for each compact subset of  $(x, z, u)$  in the domain of interest, the functions satisfy the Lipschitz inequality with a Lipschitz constant independent of  $\varsigma$  for all  $\varsigma \in \mathcal{D}$ .

The state feedback controller takes the following form

$$\dot{\theta} = \Gamma(\theta, x, w, \varsigma) \quad (5)$$

$$u = \gamma(\theta, x, w, \varsigma) \quad (6)$$

and meets the requirements listed in Assumption 2.

*Assumption 2:*

- $\Gamma$  and  $\gamma$  are locally Lipschitz functions in their arguments, uniformly in  $\varsigma$ , over the domain of interest;
- $\Gamma$  and  $\gamma$  are globally bounded functions of  $x$ .

Let the closed-loop system (1)–(4) under the state feedback controller (5)–(6) be denoted as

$$\dot{\chi} = f_r(\chi, \varsigma), \quad (7)$$

where

$$\chi = \begin{bmatrix} x \\ z \\ \theta \end{bmatrix} \in \mathbb{R}^N \quad \text{and} \quad f_r(\chi, \varsigma) = \begin{bmatrix} Ax + B\phi(x, z, \varsigma, \gamma) \\ \psi(x, z, \varsigma, \gamma) \\ \Gamma(\theta, x, w, \varsigma) \end{bmatrix}.$$

*Assumption 3:*

- The closed-loop system (7) is globally uniformly asymptotically stable with respect to a compact positively invariant set  $\mathcal{A}$ , uniformly in  $\varsigma$ ;
- $\phi(x, z, \varsigma, \gamma)$  is zero in  $\mathcal{A}$ , uniformly in  $\varsigma$ .

Instead of stabilizing an equilibrium point of the system, the problem is posed as rendering a certain compact set positively invariant and asymptotically attractive. The allure of formulating the results in this fashion, is that the separation principle is no longer limited to stabilization of an equilibrium point [17]. Some examples include servomechanisms [18], [19] and finite time convergence to a set [20]. The regulation problem for servomechanisms, for example, requires that the trajectories of the system reach an invariant manifold where the tracking error is zero, usually called the zero-error manifold. Instead of regulating the system about an equilibrium point, we desire to study the dynamics on the manifold. Therefore, the set that we wish to render positively invariant and asymptotically attractive is the zero-error manifold.

As stated in [21], uniform asymptotic stability with respect to a set  $\mathcal{A}$ , uniformly in  $\varsigma$ , stipulates the following:

- **Uniform Stability** – For each  $\epsilon > 0$  there is a  $\delta = \delta(\epsilon)$  such that

$$|\chi(t_0)|_{\mathcal{A}} \leq \delta \Rightarrow |\chi(t)|_{\mathcal{A}} < \epsilon, \quad \forall t \geq t_0 \geq 0, \quad \forall \varsigma(t) \in \mathcal{D}.$$

- **Uniform Attraction** – For each  $\epsilon > 0$  and fixed constant  $c > 0$ , independent of  $t_0$  and  $\varsigma(t)$ , there exists  $T = T(\epsilon, c)$  such that

$$|\chi(t)|_{\mathcal{A}} < \epsilon, \quad \forall t \geq t_0 + T, \quad \forall |\chi(t_0)|_{\mathcal{A}} < c, \quad \forall \varsigma(t) \in \mathcal{D}.$$

The expression  $|\chi|_{\mathcal{A}} = \inf_{\nu \in \mathcal{A}} \|\chi - \nu\|$  is the distance with respect to  $\mathcal{A}$ . Furthermore, the system is said to be globally uniformly asymptotically stable with respect to  $\mathcal{A}$ , if the uniform stability property holds with a class  $\mathcal{K}_{\infty}$  function  $\delta$ , and the uniform attraction property holds for any  $c > 0$  with  $T = T(\epsilon, c)$ . By extending the definition of stability to a compact positively invariant set instead of just an equilibrium point, a wider variety of problem formulations can be encapsulated in the above setup. Clearly, we can still address stabilization of the origin by defining  $\mathcal{A} = 0$ . However, we can just as easily structure the control objective as a regulation or tracking problem. For examples of control problems that can be formulated as stabilization with respect to a set, see [21].

### III. OBSERVER DYNAMICS

The intuition behind the nonlinear observer gain is the following:

- Achieve the desired (fast) state reconstruction with  $\varepsilon_1$  without sacrificing the steady-state performance;
- Reduce the steady-state estimation error with  $\varepsilon_2$  while maintaining an acceptable rate of convergence in the estimates.

The high-gain observer is defined as

$$\dot{\hat{x}} = A\hat{x} + B\phi_0(\hat{x}, w, \varsigma, u) + h(y - \hat{x}_1), \quad (8)$$

such that the observer duplicates the system dynamics in (2). The output feedback controller is obtained by replacing  $x$  in (5)-(6) with  $\hat{x}$  in (8). The linear-gain is

$$h_i(y - \hat{x}_1) = \alpha_i [g_1^i(y - \hat{x}_1)], \quad (9)$$

where the observer approaches the behavior of a differentiator as  $g_1 \rightarrow \infty$ . Furthermore, the order of the observer is equal to the relative degree. The nonlinear-gain is

$$h_i(y - \hat{x}_1) = \alpha_i \left[ g_1^i(y - \hat{x}_1) + d(g_2^i - g_1^i) \text{sat} \left( \frac{y - \hat{x}_1}{d} \right) \right]. \quad (10)$$

The observer gain parameters are defined as

$$g_1 = \frac{1}{\varepsilon_1} \text{ and } g_2 = \frac{1}{\varepsilon_2}, \text{ where } \varepsilon_1 < \varepsilon_2$$

and chosen to correspond to the desired transient and steady-state responses, respectively. Both  $\varepsilon_1$  and  $\varepsilon_2$  are small positive parameters. In practice, to benefit from the aid of two separate observer gains, the ratio of  $\varepsilon_2/\varepsilon_1$  should be at least 10. The parameter  $d$  is designed such that the observer gain is the smaller value  $g_2$  for  $|x_1 - \hat{x}_1| \leq d$  and  $d > \mu$ .

Notice that  $i$  is both an index (subscript) and an exponent (superscript) in (9) and (10). For instance,  $h$  is a vector, where  $h_1$  is the first element,  $h_2$  the second element, and so forth. On the other hand,  $g_1^2$  is  $g_1$  squared,  $g_1^3$  is  $g_1$  cubed, and so forth. The same can be said for  $g_2$ . The representation of the nonlinear-gain shown in (10) is derived using Proposition 1 of [22],

which states that a piecewise linear function can be written as a summation of a linear function with a saturation function. The function “sat” denotes the saturation function defined as

$$\text{sat}(e) = \begin{cases} e, & \text{if } |e| \leq 1 \\ \text{sign}(e), & \text{if } |e| > 1 \end{cases}. \quad (11)$$

The function  $\phi_0$  is a nominal model of  $\phi$ .

*Assumption 4:*

- $\phi_0$  is locally Lipschitz in its arguments, uniformly in  $\varsigma$ , over the domain of interest;
- $\phi_0$  is globally bounded in  $\hat{x}$  and zero in  $\mathcal{A}$ .

The  $\alpha_i$ 's are designed such that the roots of

$$s^n + \alpha_1 s^{n-1} + \dots + \alpha_{n-1} s + \alpha_n = 0 \quad (12)$$

are real and negative, with one fast root and  $(n-1)$  slow real roots. In this case, (12) is written as

$$(s^{n-1} + \beta_1 s^{n-2} + \dots + \beta_{n-2} s + \beta_{n-1})(s + \lambda) = 0, \quad (13)$$

where the first polynomial is Hurwitz with  $\mathcal{O}(1)$  real roots and  $\lambda \gg 1$ . In particular, we desire that the first component of the estimation error,  $x_1 - \hat{x}_1$ , decrease almost monotonically; close to a first-order equation whose solution is monotonic. This behavior is needed to ensure that the first state of the estimation error decays towards a positively invariant set, where the observer gain  $\varepsilon_2$  is active during steady-state. Before the system trajectories reach this set, the expectation is that  $\varepsilon_1$  is the active observer gain. Relating (12) to (13), it can be seen that  $\alpha_1 = \lambda + \beta_1$ ,  $\alpha_i = \beta_{i-1}\lambda + \beta_i \forall 1 < i < n$  and  $\alpha_n = \beta_{n-1}\lambda$ .

### IV. CLOSED-LOOP SYSTEM ANALYSIS

For the closed-loop system analysis, the observer dynamics are replaced by the equivalent dynamics of the scaled estimation error

$$\eta = D(\varepsilon_1)(x - \hat{x}), \quad (14)$$

where  $D(\varepsilon_1) = \text{diag}[1, \varepsilon_1, \dots, \varepsilon_1^{n-1}]$  and  $\varepsilon_1$  is a constant. The closed-loop system under the output feedback controller can be written as  $\dot{\chi} = f(\chi, \varsigma, D^{-1}(\varepsilon_1)\eta)$

$$\begin{aligned} \dot{\chi} &= \begin{bmatrix} Ax + B\phi(x, z, \varsigma, \gamma(\theta, x - D^{-1}(\varepsilon_1)\eta, w, \varsigma)) \\ \psi(x, z, \varsigma, \gamma(\theta, x - D^{-1}(\varepsilon_1)\eta, w, \varsigma)) \\ \Gamma(\theta, x - D^{-1}(\varepsilon_1)\eta, w, \varsigma) \end{bmatrix} \\ \varepsilon_1 \dot{\eta} &= A_0 \eta + B_0 v + \varepsilon_1^n B \delta(\chi, \varsigma, w, D^{-1}(\varepsilon_1)\eta) + \bar{h} \bar{\delta}(\eta_1 + v), \end{aligned} \quad (15)$$

where

$$A_0 = \begin{bmatrix} -\alpha_1 & 1 & \dots & \dots & 0 \\ -\alpha_2 & 0 & 1 & \dots & 0 \\ \vdots & & & \ddots & \vdots \\ -\alpha_{n-1} & \dots & \dots & 0 & 1 \\ -\alpha_n & \dots & \dots & \dots & 0 \end{bmatrix}, \quad B_0 = \begin{bmatrix} -\alpha_1 \\ -\alpha_2 \\ \vdots \\ -\alpha_{n-1} \\ -\alpha_n \end{bmatrix},$$

$$\bar{h}_i = \alpha_i \left[ 1 - \left( \frac{\varepsilon_1}{\varepsilon_2} \right)^i \right] \text{ and } \bar{\delta}(\eta_1 + v) = \text{sat} \left( \frac{\eta_1 + v}{d} \right) d.$$

The matrix  $A_0$  is Hurwitz, and  $|\bar{\delta}| \leq d$ . The function  $\delta(\chi, \varsigma, w, D^{-1}(\varepsilon_1)\eta)$  is defined as  $\phi(x, z, \varsigma, \gamma(\theta, \hat{x}, w, \varsigma)) -$

$\phi_0(\hat{x}, w, \varsigma, \gamma(\theta, \hat{x}, w, \varsigma))$ . Notice that  $\varepsilon_1^n$  is a coefficient of the term  $\delta$ . Therefore, the error resulting from the model uncertainty approaches 0 as  $\varepsilon_1 \rightarrow 0$ . The system (15)-(16) resembles a model appearing in the standard singularly perturbed form, as shown in [23]. The primary difference between this system and the standard form is the presence of the negative powers of  $\varepsilon_1$  in the term  $D^{-1}(\varepsilon_1)\eta$ . However,  $\delta$  is a globally bounded function in  $\hat{x}$ , implying that it is also globally bounded in  $D^{-1}(\varepsilon_1)\eta$ . This property allows us to extend the analysis associated with standard singularly perturbed systems to the case involving (15)-(16). The slow dynamics of (15) can be approximated in the absence of measurement noise ( $v = 0$ ) by setting  $\varepsilon_1 = 0$ , which yields  $\eta = 0$ . This reduces (15) to the closed-loop system (7) under the state feedback controller (5)-(6). Moreover, the system is globally uniformly asymptotically stable with respect to the compact positively invariant set  $\mathcal{A}$ . Then, according to a converse Lyapunov theorem in [24], there exists a smooth Lyapunov function  $V(\chi)$  and positive definite functions  $U_1$ ,  $U_2$ , and  $U_3$ , with respect to the set  $\mathcal{A}$ , such that

$$a_1(\|\chi\|_{\mathcal{A}}) \leq U_1(\chi) \leq V(\chi) \leq U_2(\chi) \leq a_1(\|\chi\|_{\mathcal{A}}) \quad (17)$$

$$\frac{\partial V}{\partial \chi} f(\chi, \varsigma, 0) \leq -U_3(\chi) \leq a_2(\|\chi\|_{\mathcal{A}}) \quad (18)$$

for all  $\varsigma \in \mathcal{D}$ , where  $a_1(\|\chi\|_{\mathcal{A}})$  is a class  $\mathcal{K}_\infty$  function and  $a_2(\|\chi\|_{\mathcal{A}})$  is a class  $\mathcal{K}$  function.

**Theorem:** Let Assumptions 1 through 4 hold and consider the closed-loop system (7) with the observer (8). Moreover, let  $\mathcal{M}$  be any compact set in  $\mathbb{R}^N$  and  $\mathcal{N}$  be any compact subset of  $\mathbb{R}^n$ , where  $\chi(t_0) \in \mathcal{M}$  and  $\hat{x}(t_0) \in \mathcal{N}$ . Then, given the positive constant  $\varepsilon_f = \varepsilon_1/\varepsilon_2 < 1$ , there is a positive constant  $\lambda^*$  such that for  $\lambda > \lambda^*$ , the following properties hold:

- There exist positive constants  $\mu^*$  and  $c_a$  such that for  $\mu < \mu^*$  there is a constant  $\varepsilon_a = \varepsilon_a(\mu) > c_a\mu^{1/n}$  with  $\lim_{\mu \rightarrow 0} \varepsilon_a(\mu) = \varepsilon_a^* > 0$ , such that for each  $\varepsilon_2 \in (c_a\mu^{1/n}, \varepsilon_a]$  the trajectories of the closed-loop system are bounded for all  $t \geq 0$ .
- There exists  $\mu_2^* > 0$  and a class  $\mathcal{K}$  function  $\rho_1$  such that for every  $\mu < \mu_2^*$  and every  $\Upsilon_1 > \rho_1(\mu)$ , there are constants  $T_\Upsilon = T_\Upsilon(\Upsilon_1) \geq 0$  and  $\varepsilon_b = \varepsilon_b(\mu, \Upsilon_1) > c_a\mu^{1/n}$ , with  $\lim_{\mu \rightarrow 0} \varepsilon_b(\mu, \Upsilon_1) = \varepsilon_b^*(\Upsilon_1) > 0$ , such that for each  $\varepsilon_2 \in (c_a\mu^{1/n}, \varepsilon_b]$

$$\max\{\|\chi(t)\|_{\mathcal{A}}, \|x(t) - \hat{x}(t)\|\} \leq \Upsilon_1, \quad \forall t \geq T_\Upsilon. \quad (19)$$

- There exist  $\mu_3^* > 0$  and a class  $\mathcal{K}$  function  $\rho_2$  such that for every  $\mu < \mu_3^*$  and every  $\Upsilon_2 > \rho_2(\mu)$ , there is a constant  $\varepsilon_c = \varepsilon_c(\mu, \Upsilon_2) > c_a\mu^{1/n}$ , with  $\lim_{\mu \rightarrow 0} \varepsilon_c(\mu, \Upsilon_2) = \varepsilon_c^*(\Upsilon_2) > 0$ , such that for each  $\varepsilon_2 \in (c_a\mu^{1/n}, \varepsilon_c]$

$$\|\chi(t) - \chi_r(t)\| \leq \Upsilon_2, \quad \forall t \geq t_0 \quad (20)$$

where  $\chi_r(t)$  is the solution of (7) with  $\chi_r(t_0) = \chi(t_0)$ . The last two bullet items are similar to Theorem 1 of [12].

**Remark:** The procedure for choosing the design parameters  $\varepsilon_1$ ,  $\varepsilon_2$  and  $d$  is fairly simple.

- Initially, the observer gain  $\varepsilon_1$  should be chosen to correspond to the desired transient behavior in the estimation error dynamics; the approximate results can be seen by running the chosen value in a linear-gain observer. This is accomplished by testing a wide range of values for the  $\varepsilon_1$

parameter, followed by evaluating the observer and system dynamics for each scenario. Acceptable  $\varepsilon$  values will typically fall in the range of  $[0.00001, 0.1]$ . Keep in mind that the actual range will vary according to the system of interest and the bound on the measurement noise.

- Set the parameter  $\varepsilon_1$ , in the nonlinear-gain observer, to the value that best captures the desired observer and system transient behavior.
- In order to find the appropriate value of  $\varepsilon_2$ , test a wide range of values for the  $\varepsilon_2$  parameter in a linear-gain observer, and evaluate the performance as done in the  $\varepsilon_1$  case. The range of acceptable  $\varepsilon_2$  values may need to be narrowed from the previous  $\varepsilon_1$  range to prevent violating the condition on the ratio of the  $\varepsilon$ 's.
- Choose the value of  $\varepsilon_2$  that achieves the desired steady-state estimation error dynamics.
- The value of  $d$  should be chosen as small as possible, while respecting the lower bound  $d > \mu$ , imposed by the measurement noise. By choosing the value of  $d$  small, the nonlinear observer is able to maximize the use of  $\varepsilon_1$  during the transient phase; this is done to minimize the settling time.
- Rerun the system utilizing the chosen value for  $\varepsilon_1$ . Make a note of the settling time and the corresponding estimation error for  $x_2 - \hat{x}_2$ . Set  $d$  equal to that estimation error, and run the nonlinear-gain observer.
- Increase the initial value of  $d$  until the desired transient performance is achieved.

**Proof:** In order to place the set  $\mathcal{M}$  in the interior of  $\Omega_c = \{V(\chi) \leq c\} \subset \mathbb{R}^N$ , choose  $c > \max_{\chi \in \mathcal{M}} V(\chi)$ . The set  $\Omega_c$  is compact for any choice of  $c$ . There is a constant  $L_\delta > 0$ , independent of  $\varepsilon_1$ , such that  $\|\delta\| \leq L_\delta$  for all  $\chi \in \Omega_c$  and  $\eta \in \mathbb{R}^n$ . The boundness of  $\delta$  follows from the boundness of  $\phi$  for  $x$  on a compact set.

Consider the fast representation in (16) for  $\chi \in \Omega_c$ . This equation possesses both slow and fast variables due to the choice of the eigenvalues shown in (13). To transform (16) into the singularly perturbed form,  $A_0$  and  $B_0$  are represented as  $A_0 = A_{01}\lambda + A_{02}$  and  $B_0 = B_{01}\lambda + B_{02}$ , where

$$A_{01} = \begin{bmatrix} -1 & 0 & \cdots & 0 \\ -\beta_1 & 0 & \cdots & 0 \\ \vdots & & \ddots & \\ -\beta_{n-2} & \cdots & 0 & 0 \\ -\beta_{n-1} & \cdots & \cdots & 0 \end{bmatrix}, \quad B_{01} = \begin{bmatrix} -1 \\ -\beta_1 \\ \vdots \\ -\beta_{n-2} \\ -\beta_{n-1} \end{bmatrix},$$

$$A_{02} = \begin{bmatrix} -\beta_1 & 1 & \cdots & \cdots & 0 \\ -\beta_2 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ -\beta_{n-1} & \cdots & \cdots & 0 & 1 \\ 0 & \cdots & \cdots & \cdots & 0 \end{bmatrix} \quad \text{and} \quad B_{02} = \begin{bmatrix} -\beta_1 \\ -\beta_2 \\ \vdots \\ -\beta_{n-1} \\ 0 \end{bmatrix}.$$

The procedure from [23] is used with the change of coordinates

$$\begin{bmatrix} \zeta \\ \eta_1 \end{bmatrix} = T\eta,$$

$$\text{where } T = \begin{bmatrix} Y \\ Z \end{bmatrix} = \begin{bmatrix} -\beta_1 & 1 & 0 & \cdots & 0 \\ -\beta_2 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ -\beta_{n-1} & 0 & \cdots & \cdots & 1 \\ 1 & 0 & \cdots & \cdots & 0 \end{bmatrix} \quad (21)$$

and  $T^{-1} = \begin{bmatrix} M & N \end{bmatrix}$  with  $M \in \mathbb{R}^{n \times (n-1)}$  and  $N \in \mathbb{R}^{(n-1) \times n}$ . Applying this change of coordinates to (16) yields

$$\varepsilon_1 \dot{\zeta} = Y A_{02} M \zeta + Y B_{02} v + \varepsilon_1^n Y B \delta + Y \bar{h} \bar{\delta} \quad (22)$$

$$\varepsilon_1 \dot{\eta}_1 = -\lambda \eta_1 + \zeta_1 - (\lambda + \beta_1) v + (\lambda + \beta_1) \left(1 - \frac{\varepsilon_1}{\varepsilon_2}\right) \bar{\delta}, \quad (23)$$

where

$$Y \bar{h} = \begin{bmatrix} -\beta_1 & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ -\beta_{n-1} & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} (\lambda + \beta_1) \left(1 - \frac{\varepsilon_1}{\varepsilon_2}\right) \\ (\lambda \beta_1 + \beta_2) \left(1 - \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^2\right) \\ \vdots \\ \lambda \beta_{n-1} \left(1 - \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^n\right) \end{bmatrix}$$

$$= \lambda \frac{\varepsilon_1}{\varepsilon_2} a + b,$$

$$a = \begin{bmatrix} \beta_1 \left(1 - \frac{\varepsilon_1}{\varepsilon_2}\right) \\ \beta_2 \left(1 - \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^2\right) \\ \vdots \\ \beta_{n-1} \left(1 - \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^{n-1}\right) \end{bmatrix},$$

$$b = \begin{bmatrix} -\beta_1^2 \left(1 - \frac{\varepsilon_1}{\varepsilon_2}\right) + \beta_2 \left(1 - \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^2\right) \\ -\beta_1 \beta_2 \left(1 - \frac{\varepsilon_1}{\varepsilon_2}\right) + \beta_3 \left(1 - \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^3\right) \\ \vdots \\ -\beta_{n-1} \beta_1 \left(1 - \frac{\varepsilon_1}{\varepsilon_2}\right) \end{bmatrix},$$

$$Y A_{01} = 0, \quad Y B_{01} = 0, \quad Z A_{01} M = 0, \quad Z A_{01} N = -1,$$

$Z A_{02} M = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{1 \times (n-1)}$ ,  $A_{02} N = 0$ , and  $Y A_{02} M$  is by design a Hurwitz matrix. The solution matrix  $P$  to the Lyapunov equation  $P Y A_{02} M + (Y A_{02} M)^T P = -I$  is symmetric and positive definite. Let the Lyapunov function candidate for (22) be chosen as  $W_1 = \zeta^T P \zeta$ . It can be shown that

$$\dot{W}_1 \leq -\frac{1}{\varepsilon_1} \|\zeta\|^2 + \frac{2}{\varepsilon_1} \|\zeta\| \left( \|P Y B_{02}\| \mu + \varepsilon_1^n \|P Y B\| L_\delta + \|P\| \left( \frac{\varepsilon_1}{\varepsilon_2} \lambda \|a\| + \|b\| \right) d \right). \quad (24)$$

It follows that

$$\dot{W}_1 \leq -\frac{1}{2\varepsilon_1 \|P\|} W_1, \quad \forall W_1 \geq L_W \quad (25)$$

for

$$L_W = \left( 4\sqrt{\|P\|} \left( \|P Y B_{02}\| \mu + \varepsilon_1^n \|P Y B\| L_\delta + \|P\| \left( \frac{\varepsilon_1}{\varepsilon_2} \lambda \|a\| + \|b\| \right) d \right) \right)^2. \quad (26)$$

Thus,  $\zeta$  is bounded. The bound (26) is reached within the interval  $[t_0, t_0 + T_1(\varepsilon_1)]$ , where

$$T_1(\varepsilon_1) = 4\|P\| \varepsilon_1 \ln \left( \frac{k_{w_1}}{\sigma_1 \varepsilon_1^n} \right) \rightarrow 0 \quad \text{as } \varepsilon_1 \rightarrow 0 \quad (27)$$

and  $k_{w_1} > 0$  is a constant independent of  $\varepsilon_1$  and  $\varepsilon_2$ , and  $\sigma_1 = \sqrt{\|P\|} \|P Y B\| L_\delta$ .

Meanwhile, the next step is to show that the trajectories of (23) reach a positively invariant strip defined as

$$\{|x_1 - \hat{x}_1| \leq L\}, \quad (28)$$

where  $0 < L < d$ . Using the Lyapunov function  $W_2 = (1/2)\eta_1^2$ , it can be shown that

$$\dot{W}_2 = -\frac{\lambda}{\varepsilon_2} \eta_1 (\eta_1 + v) - \lambda \left( \frac{1}{\varepsilon_1} - \frac{1}{\varepsilon_2} \right) \eta_1 (\eta_1 + v - \bar{\delta}) + \eta_1 \left[ \beta_1 \left( \frac{1}{\varepsilon_1} - \frac{1}{\varepsilon_2} \right) \bar{\delta} - \frac{1}{\varepsilon_1} \beta_1 v + \frac{1}{\varepsilon_1} \zeta_1 \right]. \quad (29)$$

It follows from  $d > |v|$  that  $\text{sign}(\eta_1) = \text{sign}(\eta_1 + v)$  whenever  $|\eta_1 + v| \geq d$  and

$$-\lambda \left( \frac{1}{\varepsilon_1} - \frac{1}{\varepsilon_2} \right) \eta_1 (\eta_1 + v - \bar{\delta}) \leq 0.$$

To ensure that the condition in (28) is satisfied, we require a tighter bound on the state  $\zeta_1$  than what is provided in (25). Thus, the ultimate bound on  $|\zeta_1|$  is denoted as

$$c_0 + k_1 \mu + k_2 \varepsilon_1^n L_\delta + k_3 \frac{\varepsilon_1}{\varepsilon_2} \lambda d + k_4 d, \quad (30)$$

where  $c_0$ , a constant due to initial conditions, can be made arbitrarily small for  $t$  large enough;  $k_1$ ,  $k_2$ ,  $k_3$ , and  $k_4$  are positive constants. Using the ultimate bound in (30), after  $t_0 + T_1(\varepsilon_1)$ ,

$$\dot{W}_2 \leq -\frac{(1-\theta)\lambda}{\varepsilon_2} |\eta_1|^2, \quad \forall |\eta_1| \geq U \quad (31)$$

for

$$U = \frac{\mu}{\theta} + \frac{\varepsilon_2}{\varepsilon_1} \left( \frac{c_1 \mu + c_2 d + c_0 + k_2 \varepsilon_1^n L_\delta}{\lambda \theta} \right) + \frac{k_3 d}{\theta}, \quad (32)$$

where  $c_1 = \beta_1 + k_1$ ,  $c_2 = \beta_1 (1 - \varepsilon_1/\varepsilon_2) + k_4$  and  $\varepsilon_1 \leq 1$ . The parameter  $\theta \in (0, 1)$  will be chosen in a later step.

To ensure that (28) implies

$$|x_1 - \hat{x}_1 + v| \leq d, \quad (33)$$

$d$  is chosen as  $d > L + \mu$ . In order for the strip (28) to be positively invariant, we require the choice of  $L$  to be greater than  $U$ . This requirement leads to the inequality  $U < L < d - \mu$ . We rewrite the foregoing inequality as  $U + \mu < d$ , and revisit  $U$  defined in (32). Recall that the terms in  $U$  containing  $\mu$  in the numerator and/or  $\lambda$  in the denominator can be made sufficiently small. However, the third term,  $k_3 d/\theta$ , does not immediately appear to be necessarily small. To ensure that the inequality  $U + \mu < d$  is satisfied, the constant  $k_3$  needs to be handled with care. In particular, we need  $k_3/\theta < 1$  to avoid violating the foregoing inequality. According to Lemma 1 of [11], the constant  $k_3$  arises

from the response of the system  $\varepsilon_1 \dot{\zeta} = Y A_{02} M \bar{\zeta} + a \lambda (\varepsilon_1 / \varepsilon_2) \bar{\delta}$  and is given by

$$k_3 = \int_0^\infty |\bar{C} e^{Y A_{02} M t} a| dt \quad (34)$$

for  $\bar{C} = [1 \ 0 \ \dots \ 0]$ . Define

$$G(s) = \bar{C}(sI - Y A_{02} M)^{-1} a. \quad (35)$$

Then, by selecting the poles and zeros of  $G(s)$  to be real and distinct such that

$$G(s) = K \left( \prod_{i=1}^{\bar{m}} \frac{s + \bar{z}_i}{s + \bar{p}_i} \right) \left( \prod_{j=\bar{m}+1}^{\bar{n}} \frac{1}{s + \bar{p}_j} \right),$$

where  $\bar{m} \leq \bar{n}$ ,  $\bar{z}_i > \bar{p}_i$  for  $i = 1, \dots, \bar{m}$ , the impulse response of (34) is nonnegative; see [25] for the proof. This assumption can always be satisfied by an appropriate choice of  $\beta_1, \dots, \beta_{n-1}$  and  $\varepsilon_1 / \varepsilon_2$ ; see [26, Appendix A] for examples. Hence,

$$k_3 = \int_0^\infty \bar{C} e^{Y A_{02} M t} a dt = G(s) \Big|_{s=0} \quad (36)$$

where solving for  $k_3$  leads to

$$Y A_{02} M = \begin{bmatrix} -\beta_1 & 1 & 0 & \dots & 0 \\ -\beta_2 & 0 & 1 & \dots & 0 \\ \vdots & & & \ddots & \vdots \\ -\beta_{n-2} & 0 & \dots & \dots & 1 \\ -\beta_{n-1} & 0 & \dots & \dots & 0 \end{bmatrix},$$

$$\begin{aligned} & \bar{C}(sI - Y A_{02} M)^{-1} a \\ &= \frac{\beta_1 \left(1 - \frac{\varepsilon_1}{\varepsilon_2}\right) s^{n-2} + \dots + \beta_{n-1} \left(1 - \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^{n-1}\right)}{s^{n-1} + \beta_1 s^{n-2} + \dots + \beta_{n-1}} \end{aligned}$$

and  $k_3 = 1 - (\varepsilon_1 / \varepsilon_2)^{n-1}$ . Therefore,

$$U + \mu = \frac{\mu}{\theta} + \mu + \frac{c_1 \mu + c_2 d + c_0 + k_2 \varepsilon_1^n L \delta}{\varepsilon_f \lambda \theta} + \frac{(1 - \varepsilon_f^{n-1}) d}{\theta}.$$

Recall that  $\mu$  must be small enough and  $\lambda$  chosen large enough to ensure that  $U + \mu < d$ . Let

$$\mu_1^* = \frac{(1 - k) \varepsilon_f^{n-1} d}{2 - k \varepsilon_f^{n-1}}, \quad \lambda^* = \frac{c_1 \mu_1^* + c_2 d + k_2 L \delta}{\varepsilon_f^n [(1 - k) d + k \mu_1^*] - 2 \varepsilon_f \mu_1^*}$$

and require that  $\mu \in (0, \mu_1^*)$ ,  $\lambda > \lambda^*$  and  $k \in (0, 1)$ . Substituting the upper bound  $\mu_1^*$  and the lower bound  $\lambda^*$  into the expression for  $U + \mu$  leads to

$$\begin{aligned} & U + \mu \\ &< \frac{\mu_1^*}{\theta} + \mu_1^* + \frac{1}{\theta} [\varepsilon_f^{n-1} (1 - k) d + \varepsilon_f^{n-1} k \mu_1^* - 2 \mu_1^*] + \frac{(1 - \varepsilon_f^{n-1}) d}{\theta} \\ &= \frac{1}{\theta} [(1 + \theta + \varepsilon_f^{n-1} k - 2) \mu_1^* + (\varepsilon_f^{n-1} (1 - k) + 1 - \varepsilon_f^{n-1}) d] \\ &= \frac{1}{\theta} [(1 + \theta + \varepsilon_f^{n-1} k - 2) \mu_1^* + (1 - \varepsilon_f^{n-1} k) d]. \end{aligned}$$

By choosing  $\theta = 1 - k \varepsilon_f^{n-1}$  and  $\varepsilon_1 \leq 1$ , the condition  $U + \mu < d$  is satisfied.

We will now show that all trajectories reach the strip (28) in finite time. The following inequality originates from (31)

$$\dot{W}_2 \leq -\lambda \frac{2k \varepsilon_f^n}{\varepsilon_1} W_2, \quad \forall W_2 \geq \frac{1}{2} L^2. \quad (37)$$

Therefore, the set

$$\Sigma_2 = \left\{ W_2 \leq \frac{1}{2} L^2 \right\} = \{ |\eta_1| \leq L \} \quad (38)$$

is positively invariant. If  $\eta_1(t_0)$  is outside of  $\Sigma_2$ , then from (37)

$$W_2(\eta_1(t)) \leq W_2(\eta_1(t_0)) \exp \left( -\lambda \frac{2k \varepsilon_f^n}{\varepsilon_1} (t - t_0) \right). \quad (39)$$

From the scaling (14), it can be seen that whenever  $x(t_0)$  and  $\hat{x}(t_0)$  are bounded, there exists a constant  $k_{w_2} > 0$ , independent of  $\varepsilon_1$  and  $\varepsilon_2$ , such that  $W_2(\eta_1(t_0)) \leq k_{w_2}^2$ . From (38) and (39), it can be seen that  $\eta_1$  reaches the set  $\Sigma_2$  within the time interval  $[t_0, t_0 + T_2(\varepsilon_1)]$ , where

$$T_2(\varepsilon_1) = \frac{\varepsilon_1}{\lambda k \varepsilon_f^n} \ln \left( \frac{k_{w_2}}{\sqrt{\frac{1}{2}} L} \right) \rightarrow 0 \text{ as } \varepsilon_1 \rightarrow 0. \quad (40)$$

At this point,  $\eta_1$  is inside the strip (28) and cannot leave the strip for all future time. Inside the strip, the parameter  $\varepsilon_2$  is driving the dynamics of the high-gain observer, not  $\varepsilon_1$ . Therefore, it is appropriate to alter the scaling (14) to obtain

$$\xi = D(\varepsilon_2)(x - \hat{x}), \quad (41)$$

where  $D(\varepsilon_2) = \text{diag}[1, \varepsilon_2, \dots, \varepsilon_2^{n-1}]$ . Then, the error dynamics become

$$\varepsilon_2 \dot{\xi} = A_0 \xi + \varepsilon_2^n B \delta + B_0 v, \quad (42)$$

which is valid for trajectories inside the strip. Take the Lyapunov function candidate as  $W_3 = \xi^T S \xi$ , where  $S$  is the positive definite symmetric solution to the Lyapunov equation  $S A_0 + A_0^T S = -I$ . Then, it can be shown that

$$\dot{W}_3 \leq -\frac{1}{2 \varepsilon_2 \|S\|} W_3, \quad \forall W_3 \geq (\sigma_2 \varepsilon_2^n + \sigma_3 \mu)^2, \quad (43)$$

where  $\sigma_2 = 4 \|SB\| L \delta \sqrt{\|S\|}$  and  $\sigma_3 = 4 \|SB_0\| \sqrt{\|S\|}$ . Therefore, the set

$$\Sigma_3 = \{ W_3 \leq (\sigma_2 \varepsilon_2^n + \sigma_3 \mu)^2 \} \quad (44)$$

is positively invariant. For  $\xi(t_0)$  outside  $\Sigma_3$ , it can be seen from (43) that

$$W_3(\xi(t)) \leq W_3(\xi(t_0)) \exp \left( \frac{-(t - t_0)}{2 \varepsilon_2 \|S\|} \right). \quad (45)$$

From the scaling equation in (41), for bounded  $x(t_0)$ ,  $\hat{x}(t_0)$  and  $\varepsilon_2$ , there exists a constant  $k_{w_3} > 0$ , independent of  $\varepsilon_2$ , such that  $W_3(\xi(t_0)) \leq k_{w_3}^2$ . It follows from (44) and (45) that  $\xi$  reaches the set  $\Sigma_3$  within the time interval  $[t_0, t_0 + T_3(\varepsilon_2)]$ , where

$$T_3(\varepsilon_2) = 4 \varepsilon_2 \|S\| \ln \left( \frac{k_{w_3}}{\sigma_2 \varepsilon_2^n} \right) \rightarrow 0 \text{ as } \varepsilon_2 \rightarrow 0. \quad (46)$$

Inside the set  $\Sigma_3$

$$\|x(t) - \hat{x}(t)\| = \|D^{-1}(\varepsilon_2)\xi(t)\| \leq \varepsilon_2\gamma_1 + \frac{\mu}{\varepsilon_2^{n-1}}\gamma_2 \triangleq F_r(\varepsilon_2, \mu) \quad (47)$$

for  $\gamma_1 = \sigma_2/\sqrt{\lambda_{\min}(S)}$  and  $\gamma_2 = \sigma_3/\sqrt{\lambda_{\min}(S)}$ . Therefore, all of the trajectories are traveling towards the positively invariant set  $\Sigma = \{\Sigma_2 \cap \Sigma_3\}$ . Moreover, the set  $\Sigma$  is reached within the time interval

$$T(\varepsilon_1, \varepsilon_2) = T_1(\varepsilon_1) + T_2(\varepsilon_1) + T_3(\varepsilon_2) \rightarrow 0 \text{ as } \varepsilon_2 \rightarrow 0, \quad (48)$$

where the times are defined in (27), (40) and (46). Recall that  $\varepsilon_1 < \varepsilon_2$ , meaning that decreasing  $\varepsilon_2$  will eventually correspond to a decrease in  $\varepsilon_1$ . Hence, only the reduction of  $\varepsilon_2$  is explicitly listed in (48). From a practical standpoint, an advantage over the linear-gain high-gain observer is that we are no longer limited to decreasing only one parameter to minimize the time  $T(\varepsilon_1, \varepsilon_2)$ . As suggested in (48), both  $\varepsilon_1$  and  $\varepsilon_2$  can be chosen to achieve the desired convergence time. The times in (27) and (40) decrease as  $\varepsilon_1 \rightarrow 0$ , whereas (46) depends solely on the parameter  $\varepsilon_2$ . Moreover, given that  $\varepsilon_1$  is strictly less than  $\varepsilon_2$ , the total time shown in (48) is decreased as  $\varepsilon_2$  is made smaller.

We will now address the remaining arguments concerning the boundness of all trajectories, ultimate boundness where the trajectories come close to the set  $\mathcal{A} \times \{x - \hat{x} = 0\}$  and closeness of trajectories. A similar approach for the slow states  $\chi$  can be found in [12].

For  $(\chi, \eta) \in \Omega_c \times \Sigma$  the vector  $\chi$  can be represented as

$$\dot{\chi} = f(\chi, \varsigma, D^{-1}(\varepsilon_2)\xi). \quad (49)$$

The function  $f$  is globally bounded in  $D^{-1}(\varepsilon_2)\xi$ , because it is globally bounded in  $\hat{x}$  and the term  $D^{-1}(\varepsilon_2)\xi$  results from substituting  $x - D^{-1}(\varepsilon_2)\xi$  for  $\hat{x}$ . Hence, there exists a positive constant  $k_f$ , independent of  $\varepsilon_2$ , such that

$$\|f(\chi, \varsigma, D^{-1}(\varepsilon_2)\xi)\| \leq k_f. \quad (50)$$

The following lemma originally appeared in [12] and is included here for convenience; see [12] for the proof.

**Lemma 1:** The function  $F_r(\varepsilon_2, \mu)$  has the following properties for  $\varepsilon_2 > 0$  and  $\mu \geq 0$ . First,  $F_r(\varepsilon_2, \mu)$  has a global minimum at  $\varepsilon_2 = [(n-1)\gamma_2\mu/\gamma_1]^{1/n} \triangleq c_a\mu^{1/n}$  and

$$\min_{\varepsilon_2 > 0} F_r(\varepsilon_2, \mu) = \left(c_1c_a + \frac{\gamma_2}{c_a^{n-1}}\right)\mu^{1/n} \triangleq k_a\mu^{1/n}.$$

For  $\varepsilon_2 > c_a\mu^{1/n}$ ,  $F_r(\varepsilon_2, \mu)$  is a strictly increasing function of  $\varepsilon_2$  and  $F_r(\varepsilon_2, \mu) \leq k_b\varepsilon_2$ , where  $k_b = \gamma_1 + \gamma_2/c_a^n$ . Then, given  $k_r > 0$ , for every  $\mu \in [0, (k_r/k_a)^n]$  there exist  $\varepsilon_m = \varepsilon_m(\mu, k_r) \geq 0$  and  $\varepsilon_M = \varepsilon_M(\mu, k_r) > c_a\mu^{1/n}$ , with  $\varepsilon_m \leq \min\{c_a\mu^{1/n}, (\mu\gamma_2n/k_r)^{1/(n-1)}\}$  and  $\lim_{\mu \rightarrow 0} \varepsilon_M(\mu, k_r) = k_r/\gamma_1$ , such that  $F_r(\varepsilon_2, \mu) \leq k_r$  for all  $\varepsilon_2 \in (\varepsilon_m, \varepsilon_M]$ .

With  $\eta = 0$ , the output feedback expression in (15) reduces to the state feedback representation shown in (7). Then, for  $\varepsilon_2 \in (c_a\mu^{1/n}, \varepsilon_2^*)$  and  $\mu < \mu_1^*$ , there is a positive constant  $L_1$ , independent of  $\varepsilon_2$ , such that the following Lipschitz condition is satisfied

$$\|f(\chi, \varsigma, D^{-1}(\varepsilon_2)\xi) - f(\chi, \varsigma, 0)\| \leq L_1\|D^{-1}(\varepsilon_2)\xi\| \quad (51)$$

for all  $(\chi, \eta) \in \Omega_c \times \Sigma$ , where the lower bound on  $\varepsilon_2$  is chosen according to Lemma 1. Taking the derivative of the smooth Lyapunov function  $V(\chi)$  yields

$$\begin{aligned} \dot{V} &= \frac{\partial V}{\partial \chi} f(\chi, \varsigma, D^{-1}(\varepsilon_2)\xi) + \frac{\partial V}{\partial \chi} f(\chi, \varsigma, 0) - \frac{\partial V}{\partial \chi} f(\chi, \varsigma, 0) \\ &= \frac{\partial V}{\partial \chi} f(\chi, \varsigma, 0) + \frac{\partial V}{\partial \chi} [f(\chi, \varsigma, D^{-1}(\varepsilon_2)\xi) - f(\chi, \varsigma, 0)] \\ &\leq -U_3(\chi) + \left\| \frac{\partial V}{\partial \chi} \right\| \|f(\chi, \varsigma, D^{-1}(\varepsilon_2)\xi) - f(\chi, \varsigma, 0)\|. \end{aligned}$$

Let  $L_2$  be an upper bound on  $\|\partial V/\partial \chi\|$  over  $\Omega_c$ . Then,

$$\dot{V} \leq -U_3(\chi) + L_2L_1\|x - \hat{x}\| \triangleq -U_3(\chi) + L_rF_r(\varepsilon_2, \mu) \quad (52)$$

for all  $(\chi, \eta) \in \Omega_c \times \Sigma$ . Let  $L_3 = (1/L_r) \min_{\chi \in \partial\Omega_c} U_3(\chi)$  and apply Lemma 1 with  $k_r = L_3$  and set  $\mu_4^* = (k_r/k_a)^n$ . Then, for  $\mu < \mu_4^*$  and  $\varepsilon_2 \in (c_a\mu^{1/n}, \varepsilon_M]$ , we have  $\dot{V} \leq 0$  for all  $(\chi, \eta) \in \partial\Omega_c \times \Sigma$ . From the previous analysis, we have that  $\dot{W}_3 \leq 0$  for all  $(\chi, \eta) \in \Omega_c \times \partial\Sigma$ . Therefore, the set  $\Omega_c \times \Sigma$  is positively invariant.

Given  $\chi(t_0)$  is in the interior of  $\Omega_c$ , we have

$$\chi(t) - \chi(t_0) = \int_{t_0}^t f(\chi(\tau), u(\tau)) d\tau.$$

Considering that  $f$  is continuous and its arguments are bounded,  $\|f(\chi, u)\| \leq k_1$  for all  $\chi(t) \in \Omega_c$ , where  $k_1$  is independent of  $\varepsilon_2$ . Hence,

$$\|\chi(t) - \chi(t_0)\| \leq k_1(t - t_0) \quad (53)$$

as long as  $\chi(t) \in \Omega_c$ . Recall that  $\chi(t_0) \in \mathcal{M} \subset \Omega_c$ . Therefore, there exists a finite time  $T^*$ , independent of  $\varepsilon_2$ , such that  $\chi(t) \in \Omega_c$  for all  $t \in [t_0, t_0 + T^*]$ . The previous analysis showed that  $\eta$  enters the set  $\Sigma$  during the finite time period  $[t_0, t_0 + T(\varepsilon_1, \varepsilon_2)]$ , where  $T(\varepsilon_1, \varepsilon_2) \rightarrow 0$  as  $\varepsilon_2 \rightarrow 0$ . Thus, there exists  $\varepsilon_u$  such that for all  $0 < \varepsilon_2 \leq \varepsilon_u$ ,  $T(\varepsilon_1, \varepsilon_2) \leq T^*$ . Therefore, choosing  $\mu_5^*$  small enough that  $c_a(\mu_5^*)^{1/n} < \varepsilon_u$  and setting  $\mu^* = \min\{\mu_1^*, \mu_4^*, \mu_5^*\}$  and  $\varepsilon_a = \min\{\varepsilon_M, \varepsilon_u\}$ , we conclude that for  $0 \leq \mu < \mu^*$  and  $c_a\mu^{1/n} < \varepsilon_2 \leq \varepsilon_a$ , the trajectory  $(\chi, \eta)$  enters the set  $\Omega_c \times \Sigma$  during the finite time period  $[t_0, t_0 + T(\varepsilon_1, \varepsilon_2)]$  and stays there for the remainder of time. Prior to entering the set,  $\chi(t)$  and  $\eta(t)$  are bounded by (53) and (45), respectively. Therefore, the closed-loop trajectories are bounded.

**1) Ultimate Boundness:** The proof for ultimate boundness utilizes the dynamics of the  $\xi$  vector inside the strip (28).

Apply Lemma 1 to the expressions (47) and (52) by setting  $k_r = \tau$ , where  $\tau \in (k_a\mu^{1/n}, L_3]$  and set  $\varepsilon_b = \min\{\varepsilon_a, \varepsilon_M\}$ ; the (47) and (52) are satisfied for all  $t \geq t_0 + T^*$ . Thus, for  $\varepsilon_2 \in (c_a\mu^{1/n}, \varepsilon_b]$  the function  $F_r(\varepsilon_2, \mu) \leq \tau$ . Then, from (52)

$$\dot{V} \leq -U_3(\chi) + L_r\tau = -\frac{1}{2}U_3(\chi) - \frac{1}{2}U_3(\chi) + L_r\tau.$$

This results in

$$\dot{V} \leq -\frac{1}{2}U_3(\chi)$$

for all  $\chi \notin \{U_3(\chi) \leq 2L_r\tau\}$ . Given  $U_3(\chi)$  is positive definite and continuous, there is a positive constant  $\tau^* < L_3$  such that

the set  $U_3(\chi) \leq 2L_r\tau$  is compact for  $\tau \leq \tau^*$ . Let  $c_o(\tau) = \max_{U_3 \leq 2L_r\tau} \{V(\chi)\}$ , then  $c_o(\tau)$  is a nondecreasing function that tends to zero as  $\tau \rightarrow 0$ . Choose a class  $\mathcal{K}$  function  $\varphi(\tau)$  such that  $\varphi \geq c_o(\tau)$ . Then, for  $\varphi(\tau) \leq V(\chi) \leq c$

$$\dot{V} \leq -\frac{1}{2}U_3(\chi).$$

Therefore, there exists a time  $T_\varphi = T_\varphi(\tau) \geq 0$  such that

$$V(\chi(t)) \leq \varphi(\tau)$$

for all  $t \geq t_0 + T^* + T_\varphi(\tau)$ . It follows from (17) that

$$|\chi(t)|_{\mathcal{A}} \leq U_1^{-1}(\varphi(\tau)) \triangleq \rho_a(\tau)$$

for all  $t \geq t_0 + T^* + T_\varphi(\tau)$ . Moreover,

$$\max\{|\chi(t)|_{\mathcal{A}}, \|\chi(t) - \hat{x}(t)\|\} \leq \max\{\tau, \rho_a(\tau)\} \triangleq \rho_b(\tau)$$

where  $\rho_b$  is a class  $\mathcal{K}_\infty$  function. Define  $\rho_1$  as  $\rho_1(\mu) = \rho_b(k_a\mu^{1/n})$  with  $\mu < \mu_2^* = \min\{\mu^*, (\tau^*/k_a)^n\}$ , where  $\rho_1$  is a class  $\mathcal{K}_\infty$ . Given  $\Upsilon_1 > \rho_1(\mu)$ , take  $\tau = \min\{\tau^*, \rho_b^{-1}(\Upsilon_1)\}$  and set  $T_\Upsilon = t_0 + T^* + T_\varphi(\tau)$  to achieve the inequality in (19).

2) *Closeness of Trajectories*: Using the fact that the closed-loop system under state feedback is uniformly asymptotically stable with respect to the set  $\mathcal{A}$  and the inequality in (19), given  $\Upsilon_2 > 2\rho_1(\mu)$ , there exists a time  $T_{\Upsilon_2} = T_{\Upsilon_2}(\Upsilon_2) > 0$ , independent of  $\varepsilon_2$ , such that

$$|\chi(t)|_{\mathcal{A}} \leq \frac{\Upsilon_2}{2} \quad (54)$$

and

$$|\chi_r(t)|_{\mathcal{A}} \leq \frac{\Upsilon_2}{2} \quad (55)$$

for all  $t \geq T_{\Upsilon_2}$  for  $\varepsilon_2 \in (c_a\mu^{1/n}, \varepsilon_b]$ . Then,

$$\|\chi(t) - \chi_r(t)\| \leq \|\chi(t) - x\| + \|\chi_r(t) - x\| \quad (56)$$

for all  $t \geq T_{\Upsilon_2}$  and  $x \in \mathcal{A}$ . Then, we take the infimum of the right-hand side of (56), over all  $x \in \mathcal{A}$ , and substitute the values in (54) and (55). This results in the following

$$\|\chi(t) - \chi_r(t)\| \leq |\chi(t)|_{\mathcal{A}} + |\chi_r(t)|_{\mathcal{A}} \leq \Upsilon_2 \quad (57)$$

for all  $t \geq T_{\Upsilon_2}$ . Furthermore, we can see from (53) that

$$\|\chi(t) - \chi(t_0)\| \leq k_1(t - t_0) \quad (58)$$

and similarly

$$\|\chi_r(t) - \chi(t_0)\| \leq k_1(t - t_0) \quad (59)$$

for all  $t \in [t_0, t_0 + T(\varepsilon_1, \varepsilon_2)]$ . Therefore,

$$\|\chi(t) - \chi_r(t)\| \leq 2k_1T(\varepsilon_1, \varepsilon_2) \quad (60)$$

for all  $t \in [t_0, t_0 + T(\varepsilon_1, \varepsilon_2)]$ . Viewing the closed-loop system under output feedback as a perturbation of the closed-loop system under state feedback and applying Theorem 3.4 of [13] over the interval  $[t_0 + T(\varepsilon_1, \varepsilon_2), T_{\Upsilon_2}]$  gives the following

$$\|\chi(t) - \chi_r(t)\| \leq 2k_1c_3T(\varepsilon_1, \varepsilon_2) + c_4F_r(\varepsilon_2, \mu) \quad (61)$$

for some constants  $c_3 \geq 1$  and  $c_4 > 0$ , independent from  $\varepsilon_1$  and  $\varepsilon_2$ . It can be seen from (60) and (61) that the time for which (61) is valid can be simplified to  $[t_0, T_{\Upsilon_2}]$ . Furthermore, it can be verified that  $T(\varepsilon_1, \varepsilon_2)$  is a class  $\mathcal{K}$  function of  $\varepsilon_2$  for  $\varepsilon_2 \leq (1/e)(k_w3/\sigma_2)^{1/n} \triangleq \bar{\varepsilon}_2$ . The inequality  $c_a(\mu_6^*)^{1/n} < \bar{\varepsilon}_2$  can be satisfied for a sufficiently small  $\mu_6^*$ . Then, for each  $\mu < \mu_6^*$  the following statement holds

$$\begin{aligned} \min_{\varepsilon_2 \in (c_a\mu^{1/n}, \bar{\varepsilon}_2]} \{2k_1c_3T(\varepsilon_1, \varepsilon_2) + c_4F_r(\varepsilon_2, \mu)\} \\ = 2k_1c_3T(\varepsilon_f c_a\mu^{1/n}, c_a\mu^{1/n}) + c_4k_a\mu^{1/n} \\ \triangleq \rho_3(\mu). \end{aligned} \quad (62)$$

Moreover, it can be shown that  $\rho_3(\mu)$  is a class  $\mathcal{K}$  function. For each  $\Upsilon_2 > \rho_3(\mu)$ , there exists  $\bar{\varepsilon}_c = \bar{\varepsilon}_c(\mu, \Upsilon_2) > c_a\mu^{1/n}$  with the  $\lim_{\mu \rightarrow 0} \bar{\varepsilon}_c(\mu, \Upsilon_2) = \bar{\varepsilon}_c^*(\Upsilon_2) > 0$  such that for all  $\varepsilon_2 \in (c_a\mu^{1/n}, \bar{\varepsilon}_c]$

$$2k_1c_3T(\varepsilon_1, \varepsilon_2) + c_4F_r(\varepsilon_2, \mu) \leq \Upsilon_2 \quad (63)$$

is satisfied. The inequality in (20) can be found by taking

$\rho_2(\mu) = \max\{2\rho_1(\mu), \rho_3(\mu)\}$ ,  $\mu_3^* = \min\{\mu_2^*, \mu_6^*\}$  and  $\varepsilon_c = \min\{\varepsilon_b, \bar{\varepsilon}_c, \bar{\varepsilon}_2\}$ , in combination with (57), (61) with the simplified time and (63).

## V. SIMULATION: AN ADAPTIVE-GAIN OBSERVER COMPARISON

The purpose of this simulation is to compare the performance of the proposed nonlinear-gain observer with the observer in [8]. Consider the following

$$\dot{x}_1 = x_2 \quad (64)$$

$$\dot{x}_2 = -x_1^3 + u \quad (65)$$

$$y = x_1 + v \quad (66)$$

where the  $x_i$ 's are the states,  $y$  the output, and  $v$  the bounded measurement noise. The measurement noise is generated using the Simulink block "Uniform Random Number", where the magnitude is limited to  $[-0.001, 0.001]$  and the sampling time is set at 0.0008 seconds. The objective of the control is to stabilize the closed-loop system (64)-(65), which is achieved with the controller  $u = -x_2$ . Following the procedure in [8], the adaptive-gain observer parameters are set at  $k_1 = 2$ ,  $k_2 = 2$ ,  $p_1 = 10$ ,  $p_2 = 1$ , and  $b = 0.49$ .

Concerning the nonlinear-gain observer, the output feedback control saturates the estimate  $\hat{x}_2$  outside  $[-10, 10]$ . The control was not saturated for the adaptive-gain observer, given the performance of the estimation error was shown to suffer. The  $\alpha_i$ 's in the high-gain observers are chosen as  $\alpha_1 = 2$  and  $\alpha_2 = 1$ . The initial conditions are  $x_2(0) = \hat{x}_1(0) = \hat{x}_2(0) = 0$  and  $x_1(0) = 2$ .

In order to choose the observer gains, we follow the design methodology detailed in the Remark found in Section IV. Fig. 1 shows the transient response of the estimation error  $x_2 - \hat{x}_2$  for linear-gain observers utilizing various values of  $\varepsilon$  in the range of  $[0.0005, 0.9]$ . Fig. 1(a) and (b) have the fastest transient response, resulting in the smallest settling times; these are our top contenders for  $\varepsilon_1$  in the nonlinear-gain observer. Fig. 2 shows a zoomed-in version of the estimation error resulting from the observers with the two smallest  $\varepsilon$  values. Clearly, we can see that



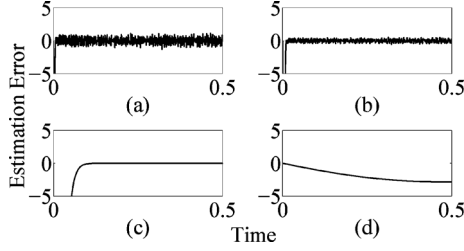


Fig. 1. Transient response of the error  $x_2 - \hat{x}_2$  vs. time for a linear-gain high-gain observer where (a)  $\varepsilon = 0.0005$ , (b)  $\varepsilon = 0.001$ , (c)  $\varepsilon = 0.01$ , and (d)  $\varepsilon = 0.9$ .

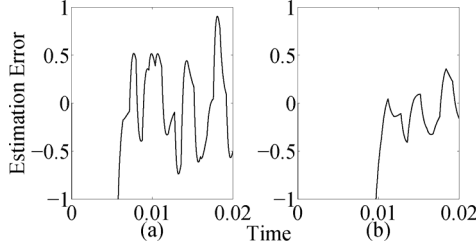


Fig. 2. Transient response of the error  $x_2 - \hat{x}_2$  vs. time for a linear-gain high-gain observer where (a)  $\varepsilon = 0.0005$  and (b)  $\varepsilon = 0.001$ .

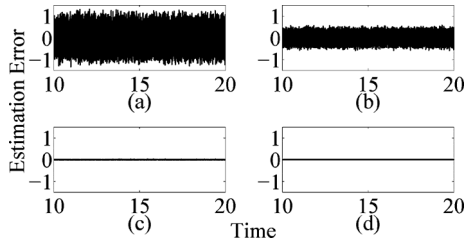


Fig. 3. Steady-state response of the error  $x_2 - \hat{x}_2$  vs. time for a linear-gain high-gain observer where (a)  $\varepsilon = 0.0005$ , (b)  $\varepsilon = 0.001$ , (c)  $\varepsilon = 0.01$ , and (d)  $\varepsilon = 0.9$ .

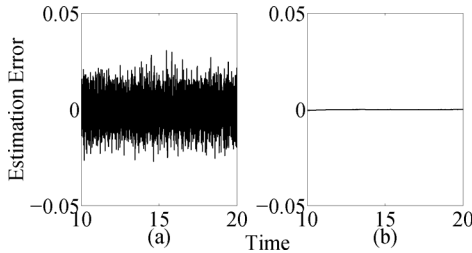


Fig. 4. Steady-state response of the error  $x_2 - \hat{x}_2$  vs. time for a linear-gain high-gain observer where (a)  $\varepsilon = 0.01$  and (b)  $\varepsilon = 0.9$ .

Fig. 2(a) has the fastest transient response when considering observers utilizing  $\varepsilon$  values in the range explored. Therefore, we will set  $\varepsilon_1 = 0.0005$  for the nonlinear-gain observer.

We will now examine the steady-state behavior of the estimation error resulting from observers with  $\varepsilon$  values in the range  $[0.0005, 0.9]$ , shown in Fig. 3. The observers with the two larger values of  $\varepsilon$  appear to produce a steady-state response of the estimation error with the least magnification of the measurement noise. Therefore, Fig. 4 takes a closer look at the estimation error for the two observers utilizing the smaller gain values. In order to achieve a smaller bound on the steady-state estimation error,  $\varepsilon_2 = 0.9$  is chosen for the nonlinear-gain observer.

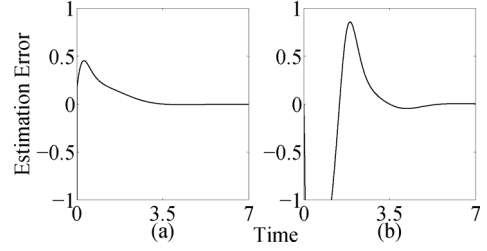


Fig. 5. Transient response of the error  $x_2 - \hat{x}_2$  vs. time for the (a) Nonlinear and (b) Adaptive gain observer.

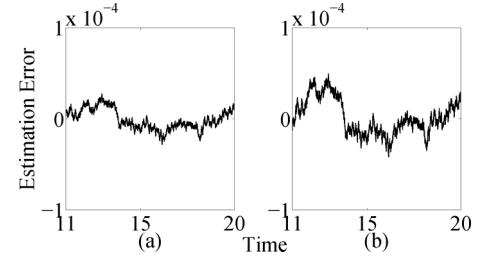


Fig. 6. Steady-state response of the error  $x_2 - \hat{x}_2$  vs. time for the (a) Nonlinear and (b) Adaptive gain observer.

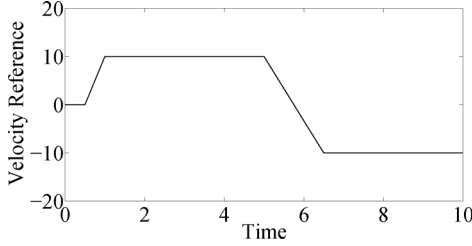
Now that we have chosen the observer gains for the nonlinear-gain observer, we must choose when each of those gains are active. Refer back to Fig. 2(a), and pay particular attention to the value of the estimation error about the settling time. The estimation error appears to settle around 0.01. However, to ensure that the estimation error reaches a positively invariant set in finite time, we set  $d = 0.15$ .

Fig. 5 provides a comparison of the transient response of the error  $x_2 - \hat{x}_2$  resulting from the nonlinear-gain and adaptive gain observers. In this case, the settling time for the two observers is fairly comparable; the estimation error resulting from the system utilizing the nonlinear-gain observer is just slightly smaller than that of the adaptive-gain observer proposed in [8]. The dynamics in steady-state of the estimation error for both the nonlinear-gain and adaptive-gain observers are shown in Fig. 6. The nonlinear-gain observer is able to achieve a smaller bound on the steady-state estimation error than the adaptive-gain observer.

Overall, it would appear as if the nonlinear-gain observer is able to outperform (if only slightly) the adaptive observer for the case presented. For the purpose of post-processing, we define the mean square error as

$$\frac{1}{n} \sum_{k=1}^n (y(k) - \hat{x}_1(k))^2, \quad (67)$$

where  $n$  is the number of samples generated by Matlab Simulink. The mean square error of  $y - \hat{x}_1$  is approximately 0.11 for the adaptive observer and 0.005 for the nonlinear-gain observer. Interestingly, if we consider the mean square error of the output in steady-state (where (67) is applied to the samples after the transient period), the nonlinear-gain observer has a mean square error of approximately  $3.7e - 7$ , and the adaptive observer  $3.4e - 7$ . Hence, the primary benefit of the nonlinear-gain observer over the adaptive-gain observer is in the simplicity of the implementation. Moreover, this simulation

Fig. 7. Velocity reference trajectory ( $\dot{r}$ ).

is not intended to suggest that the nonlinear-gain observer will always be the better choice; rather, the application should drive the type of observer implemented.

## VI. SIMULATION: FIELD CONTROLLED DC MOTOR

The system under consideration is a field controlled DC motor [13], where it is desired that the shaft angular velocity track a reference signal as shown in Fig. 7. The system is represented as

$$\dot{x}_1 = x_2 \quad (68)$$

$$\dot{x}_2 = \phi(x, z, u) \quad (69)$$

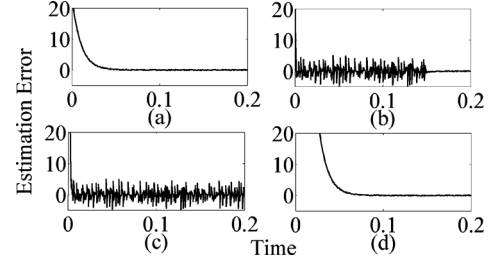
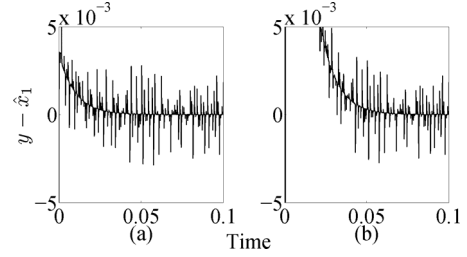
$$\dot{z} = \psi(x, z, u) \quad (70)$$

$$y = x_1 + v \quad (71)$$

$$w = z, \quad (72)$$

where  $x_1$  is the rotor position,  $x_2$  the rotor angular velocity and  $z$  the armature current. Notice that the available measurements are the rotor position and the current. Moreover, the current measurement is not an output resulting from the chain of integrators. For further details on the role of  $w$ , see Section II.

The field current is used as the source of control and is denoted by  $u$ . The controller expression is  $u = (10/w)(0.11\hat{x}_2 + \ddot{r} - 100(y - r) - 20(\hat{x}_2 - \dot{r}))$ , where feedback linearization is applied. The functions above are defined as  $\phi(x, z, u) = -0.1x_2 + 0.1zu$  and  $\psi(x, z, u) = -2z - 0.2x_2u + 200$ . The estimate  $\hat{x}_2$  is saturated outside  $[-100, 100]$ . Moreover, the controller is globally bounded in  $\hat{x}_2$ ; that is, bounded for any value of  $\hat{x}_2$  when all other variables are bounded. The saturation values are chosen such that the saturation is never active when the system is under state feedback control. The nominal value for  $\phi$  used in the observer is  $\phi_0(\hat{x}, w, u) = -0.11x_2 + 0.1wu$ . The state  $z$  is measured and need not be estimated; thus, the observer is second-order. The gains for the observers are chosen as  $\varepsilon_1 = 0.0005$  and  $\varepsilon_2 = 0.01$ . The remaining parameters are chosen as  $\alpha_1 = 71$  and  $\alpha_2 = 70$ . The initial conditions are set at  $x_1(0) = x_2(0) = \hat{x}_2(0) = 0$ ,  $\hat{x}_1(0) = 0.02$  and  $z(0) = 100$  to match values consistent with the physical system, and deliberately chosen to be unequal to ensure peaking in the transient response of the system, lending itself to a more realistic scenario. The measurement noise  $v$  is generated using the Simulink block “Uniform Random Number”, where the magnitude is limited to  $[-0.0016, 0.0016]$  and the sampling time is set at 0.0008 seconds. The noise magnitude is based on a 1000 c/r encoder. The value for the switched observer threshold is 0.05, and for the nonlinear-gain observer  $d = 0.0035$ . For the switched observer,

Fig. 8. Transient response of the error  $x_2 - \hat{x}_2$  vs. time for a (a) Nonlinear, (b) Switched, (c) Linear  $\varepsilon_1$  and (d) Linear  $\varepsilon_2$  gain high-gain observer.Fig. 9. Transient response of the innovation process  $y - \hat{x}_1$  vs. time for a (a) Nonlinear and (b) Linear  $\varepsilon_2$  gain high-gain observer.

it was shown in [12] that if the system switches before the transient response of the estimates of the higher order derivatives has subsided and entered a positively invariant set, the other transients can take  $y - \hat{x}_1$  out of the strip and subsequently cause the value of  $\varepsilon$  to switch. If this occurs, the system could be susceptible to multiple switching until all of the trajectories recover from peaking. Thus, the switched observer requires the additional component of a switching timer, based on the peaking period, that prevents the observer from switching before the trajectories of the estimation error have reached a positively invariant set. The delay timer is set for 0.15 seconds; details on how to choose this value can be found in [12].

Fig. 8 shows the transient response of the error  $x_2 - \hat{x}_2$ . As expected, the switched observer captures the behavior of the linear observer with the parameter  $\varepsilon_1$ . Unlike the switched observer, the nonlinear-gain observer does not wait for the transients to subside in both states before entering the strip. Therefore, the system utilizing the nonlinear-gain observer does not perfectly mimic the transient response of the system with the linear observer shown in Fig. 8(c). As a result, the presence of noise is more noticeable in the estimation error generated with the switched observer than with the nonlinear-gain observer. Thus, the nonlinear-gain observer is able to recover the performance of the observer in Fig. 8(d) faster than the switched observer. Fig. 9 shows that the input to the nonlinear-gain observer has a faster transient response than the linear-gain observer with the parameter  $\varepsilon_2$ .

In Fig. 10, the steady-state behavior of the estimation error is practically identical for all observers shown, with the exception of the linear  $\varepsilon_1$  observer.

Fig. 11 shows the tracking error  $x_2 - \dot{r}$  during the transient response of the observer dynamics. The transient response resulting from the system using the nonlinear-gain observer is faster than both linear observers. As shown in Fig. 12, the

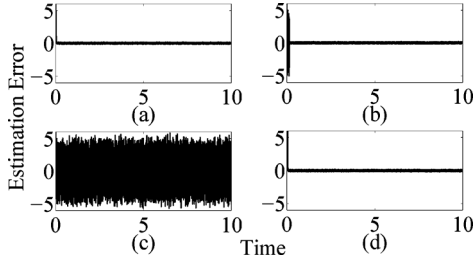


Fig. 10. Steady-state response of the error  $x_2 - \hat{x}_2$  vs. time for a (a) Nonlinear, (b) Switched, (c) Linear  $\varepsilon_1$  and (d) Linear  $\varepsilon_2$  gain high-gain observer.

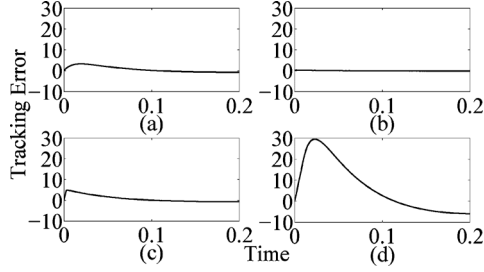


Fig. 11. Transient response of the tracking error  $x_2 - \dot{r}$  vs. time for a (a) Nonlinear, (b) Switched, (c) Linear  $\varepsilon_1$  and (d) Linear  $\varepsilon_2$  gain high-gain observer.

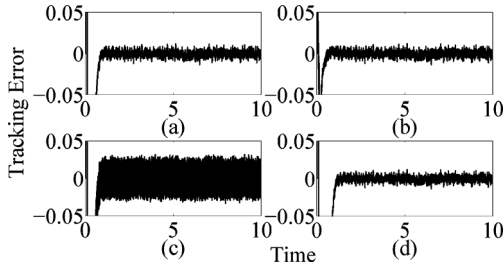


Fig. 12. Steady-state response of the tracking error  $x_2 - \dot{r}$  vs. time for a (a) Nonlinear, (b) Switched, (c) Linear  $\varepsilon_1$  and (d) Linear  $\varepsilon_2$  gain high-gain observer.

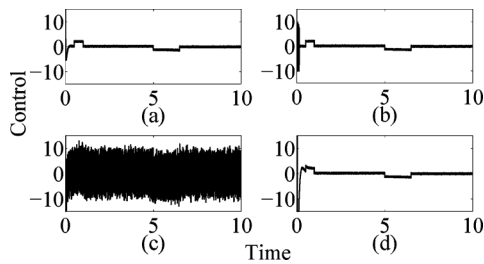


Fig. 13. Control signal for a (a) Nonlinear, (b) Switched, (c) Linear  $\varepsilon_1$  and (d) Linear  $\varepsilon_2$  gain high-gain observer.

steady-state response of the tracking errors are nearly identical for three out of the four observers.

The control effort for the four different cases is shown in Fig. 13. In particular, the nonlinear-gain observer in Fig. 13(a) and the switched-gain observer in Fig. 13(b) both appear to recover from peaking more quickly than the linear-gain observer shown in Fig. 13(d), and with greater noise immunity at steady-state than the system utilizing the linear-gain observer in Fig. 13(c). For the sake of completeness, the current  $z$  is shown for each observer implementation in Fig. 14.

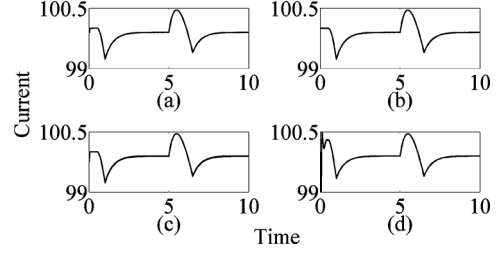


Fig. 14. Current output for a (a) Nonlinear, (b) Switched, (c) Linear  $\varepsilon_1$  and (d) Linear  $\varepsilon_2$  gain high-gain observer setup.

Overall, the nonlinear-gain observer is able to achieve better system performance than the two linear-gain observers, while bypassing the complications associated with the considered switched-gain observer.

## VII. CONCLUSIONS

When high-gain observers are utilized in the presence of measurement noise, there exists a tradeoff between fast state reconstruction and a reasonable state estimation error. The nonlinear-gain high-gain observers adequately captured the transient and steady-state performance seen in comparable linear-gain observers. Specifically, the nonlinearity was chosen to have a higher observer gain during the transient period and a lower gain afterwards, thus overcoming the tradeoff between fast state reconstruction and measurement noise attenuation. Moreover, it was assumed that all assumptions hold globally, allowing the use of a converse Lyapunov theorem of [24] in the proof. By a slight modification of the proof, it is also possible to require the assumptions to hold only in a given region of the state space and invoke the converse Lyapunov theorem of [27], which is a regional version of the theorem of [24]. In particular, the first derivative of the exogenous signals,  $\dot{z}$ , is required to be bounded.

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